

PWM-LDCT: A Multi-Vendor, Multi-Site, Paired-Dose CT Benchmark Dataset for Reproducible Low-Dose Reconstruction Research

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Abstract

We present **PWM-LDCT**, a multi-vendor, multi-site, paired-dose chest- and abdominal-CT dataset designed as a permanent, reproducibility-first benchmark for low-dose CT reconstruction research. The dataset comprises **[TODO: $N_{\text{paired scans}}$]** paired full-dose / low-dose scans across at least two manufacturer platforms (**[TODO: vendor list]**) and **[TODO: N_{sites}]** clinical sites; **[TODO: $N_{\text{vendor A}}$]** patients contribute real reduced-tube-current paired acquisitions and **[TODO: $N_{\text{vendor B}}$]** contribute physically calibrated Poisson-thinned low-dose simulations alongside their full-dose scans. Each release record ships with raw sinograms, reconstructed images at four dose levels (10%, 25%, 50%, 100%), majority-vote radiologist annotations for lesion detection and segmentation tasks, and full provenance metadata under PhysioNet credentialed access. A pip-installable Python loader (`pwm_ldct_loader`) and three reproducible Docker pipelines deliver bit-identical preprocessing across machines, and a content-addressed dataset hash is registered as an immutable L3 specification on the PWM blockchain registry so leaderboard submissions can be tied to a specific frozen benchmark version. By foregrounding multi-vendor diversity, real paired low-dose acquisitions, task-level annotations, and on-chain immutability, PWM-LDCT addresses three long-standing gaps in the field: vendor-specificity of published methods, evaluation through PSNR alone rather than clinical task performance, and benchmark drift across publication cycles. The dataset is the empirical substrate for the recurring PWM Low-Dose CT Challenge at RSNA / ISBI and the validation cohort for the signal-equivalence framework ([to be confirmed, 2027a](#)).

Background & Summary

Low-dose computed tomography (CT) reconstruction has been transformed by deep learning over the past decade ([Chen et al., 2017](#); [Wang et al., 2020](#)), with successive methods reporting incremental gains in peak signal-to-noise ratio (PSNR) and structural similarity (SSIM) against published benchmarks ([AAPM, 2016](#)). Yet three structural problems prevent the field from converging on a single, citable state-of-the-art:

1. **Vendor specificity.** Most published methods are evaluated on data from one CT manufacturer, typically Siemens (via the AAPM 2016 Low-Dose CT Grand Challenge ([AAPM, 2016](#))) or a vendor-internal dataset. A method that wins on Siemens data may degrade by 2–4 dB PSNR

on GE or Canon data because of subtle differences in raw-projection geometry, vendor-specific reconstruction kernels, and noise statistics. Without a multi-vendor test split, cross-vendor robustness cannot be measured.

2. **PSNR / SSIM are not clinical endpoints.** A reconstruction that scores well on pixel-wise metrics may smooth out clinically relevant low-contrast lesions (Kim et al., 2019). Without paired task annotations (lesion bounding boxes, segmentations, diagnostic-quality labels) reviewers cannot ask the diagnostically important question: does the low-dose reconstruction still support correct clinical decisions?

3. **Benchmark drift.** As datasets evolve (new patients added, splits reshuffled, preprocessing changed), reported numbers from year n become incomparable to year $n + 1$. The community lacks a frozen, content-addressed reference version to anchor longitudinal comparison.

PWM-LDCT is designed to close all three gaps simultaneously. Multi-vendor coverage is enforced by inclusion criteria. Per-patient majority-vote radiologist annotations support a curated set of clinical tasks (lung nodule detection, liver lesion segmentation, abdominal organ quantification). The dataset version is content-addressed and registered on the PWM blockchain registry as an immutable L3 specification, so any future paper that cites `pwm_13_spec:pwm_low_dose_ct_benchmark:1.0.0` is guaranteed to reference bit-identical data.

Table 1 situates PWM-LDCT against existing public datasets. LIDC-IDRI (Armato III et al., 2011) is the field’s largest annotated chest cohort but is single-dose and single-vendor in practice. The AAPM 2016 Low-Dose CT Grand Challenge release (AAPM, 2016) is the only public source of real paired-dose acquisitions but is limited to 10 patients on a single Siemens platform. The Mayo Clinic LDCT-and-Projection-Data collection (McCollough et al., 2017) extends AAPM 2016 with raw projections but inherits its single-vendor scope. PWM-LDCT is the first multi-vendor paired-dose release at the ≥ 500 -patient scale and the first with a content-addressed benchmark hash anchored to a public registry.

Dataset	Patients	Vendors	Real paired LD	Projections	On-chain anchor
LIDC-IDRI (Armato III et al., 2011)	1,018	1 (mostly)	No	No	
AAPM 2016 (AAPM, 2016)	10	1 (Siemens)	Yes	Yes	
Mayo LDCT-PD (McCollough et al., 2017)	299	1 (Siemens)	Yes	Yes	
PWM-LDCT (ours)	[TODO: N_{patients}]	≥ 2	Yes	Yes	Yes

Table 1: Comparison of PWM-LDCT against existing public low-dose CT datasets. “Real paired LD” indicates whether the dataset contains real (not simulated) paired full-dose / low-dose acquisitions for the same patient. “On-chain anchor” indicates whether the dataset version is content-addressed on a public blockchain registry.

The dataset is open to academic and industrial researchers under PhysioNet credentialed access (HIPAA Safe Harbor compliant for de-identified scans). A reference reconstruction method (to be confirmed, 2027b), a probabilistic signal-equivalence framework (to be confirmed, 2027a), and a permanent leaderboard (to be confirmed, 2028) are released concurrently, completing the citable substrate for the PWM Low-Dose CT Challenge.

Methods

Acquisition protocol

PWM-LDCT scans were acquired prospectively at [TODO: N_{sites}] clinical sites (University of Texas Southwestern Medical Center serving as the coordinating site) between [TODO: collection start date] and [TODO: collection end date], following an IRB-approved data sharing protocol (UTSW IRB protocol [TODO: IRB number]). Inclusion criteria were:

- Adult patients (age ≥ 18) undergoing routine chest CT for lung-cancer screening or routine abdominal CT for oncology follow-up
- Informed consent obtained for the use of de-identified imaging data in research
- Acquisition on a manufacturer-platform combination not represented at $\geq 30\%$ within the cohort (enforcing vendor diversity)

For each enrolled patient a *paired* acquisition was performed:

- **Full-dose scan:** the patient’s clinically indicated tube current setting (typically 200–400 mA depending on body habitus).
- **Real low-dose scan:** an immediately subsequent acquisition at 25% of the full-dose mA, taken within the same breath-hold for chest scans and within ≤ 30 seconds for abdominal scans to minimize motion-induced misregistration. This paired acquisition was performed on [TODO: $N_{\text{vendor A}}$] patients across vendors that support rapid sequential acquisition at differing mA.
- **Simulated low-dose data:** for [TODO: $N_{\text{vendor B}}$] additional patients (and as augmentation for the real-paired cohort) we generated 10%, 25%, and 50% dose-equivalent sinograms via physically calibrated Poisson-noise insertion on the full-dose sinogram, using the forward model implemented in PWM’s `ct_radon` modality library (PWM Protocol, 2026). The simulation protocol is documented in Section *Low-dose simulation* below; we explicitly maintain both real and simulated low-dose images in the dataset so that downstream users can assess how well methods trained on simulated low-dose generalize to real low-dose.

Low-dose simulation

We follow standard practice in the low-dose CT literature (Zeng et al., 2015; Wang et al., 2020) for simulation:

$$\tilde{s}_{\text{low}}(\ell) = -\ln\left(\frac{\text{Poisson}(I_0 \cdot r \cdot e^{-s_{\text{ref}}(\ell)})}{I_0 \cdot r}\right) \quad (1)$$

where $s_{\text{ref}}(\ell)$ is the full-dose log-attenuation projection along ray ℓ , I_0 is the assumed monochromatic incident photon count (calibrated per scanner from the acquisition’s recorded tube parameters), and $r \in \{0.10, 0.25, 0.50\}$ is the dose-reduction ratio. Both polychromatic-spectrum corrections and the standard electronic-noise floor are applied per the calibration recipe in (Zeng et al., 2015). The forward operator and Poisson sampling are implemented as the open-source `pwm_core.contrib.modalities.ct_radon` package; we explicitly do *not* reinvent the forward model so that any downstream user replicating our simulation pipeline obtains bit-identical sinograms.

Reconstruction

For each (patient, dose-level) pair we provide three image-space reconstructions:

1. **Filtered back-projection (FBP)** with a Hamming-windowed Ram-Lak filter — the canonical analytical reference.
2. **Vendor standard iterative reconstruction (IR)** — the algorithm clinically deployed on the originating scanner, where the vendor exposes a research interface (Siemens *Admirer*, GE *Veo*, Canon *AIDR3D*). Where unavailable, the field is recorded as `null`.
3. **Total-variation (TV) regularized reconstruction** — the open reference iterative method used to ensure that even vendor-restricted institutions have a reproducible iterative baseline.

All reconstructions are provided alongside the raw projections so that researchers can apply their own reconstruction algorithms to the same sinograms.

Annotation protocol

Three classes of annotations are provided:

- **Lung nodule detection (chest scans).** Bounding boxes for all pulmonary nodules ≥ 3 mm in maximum diameter, annotated independently by ≥ 2 board-certified radiologists with ≥ 5 years of experience. Discordant annotations resolved by majority vote (with a third radiologist consulted if needed). Per-nodule attributes recorded: maximum diameter (mm), location (lobe + segment), texture (solid / part-solid / non-solid).
- **Liver lesion segmentation (abdominal scans).** Per-voxel segmentation masks for hepatic lesions ≥ 5 mm in maximum diameter, with the same independent multi-reader majority-vote protocol.
- **Diagnostic-quality label (per scan).** Each radiologist rated each reconstruction on a 5-point Likert scale for *diagnostic confidence* (1 = nondiagnostic, 5 = fully diagnostic), recorded per dose level so that downstream task analyses can correlate computational reconstruction quality with radiologist-perceived diagnostic utility.

Inter-rater agreement was monitored monthly throughout the annotation campaign; Cohen’s κ remained $\geq$ **[TODO: κ threshold]** for nodule detection and Dice $\geq$ **[TODO: Dice threshold]** for liver lesion segmentation between reader pairs.

De-identification and HIPAA compliance

All scans were de-identified per HIPAA Safe Harbor (45 CFR 164.514(b)(2)). The DICOM tag whitelist and the burned-in-text scrubbing recipe are published as `schema/dicom_cleaning_spec.md` in the project repository. De-identification was verified by a two-stage pipeline: (i) automated PyDICOM tag stripping against the whitelist; (ii) OCR over every reconstructed slice using Tesseract (Smith, 2007), with any positive hit (e.g., date, name, MRN) flagged for manual review and reprocessing. Every scan in the released set passed both stages with no PHI detected.

Patient-level data splits

We partition patients (not scans) into 60% train / 20% validation / 20% test, stratified by anatomical site, vendor, and dose-pair availability. A patient’s full-dose, real low-dose, and simulated low-dose scans all reside in the same split. Split assignments are deterministic given the published random seed (42) and a patient’s de-identified ID. The validation split is provided for hyperparameter tuning; the test split is reserved for final-evaluation reporting in published work.

Data Records

The dataset is distributed via PhysioNet at [\[TODO: PhysioNet URL\]](#) under credentialed access. The full release contains:

Path	Contents
patients/<patient_id>/	Per-patient directory
full_dose/	DICOM, reconstructed image, sinogram
low_dose_real/	DICOM, image, sinogram (where available)
low_dose_sim_r010/	Simulated 10% dose
low_dose_sim_r025/	Simulated 25% dose
low_dose_sim_r050/	Simulated 50% dose
annotations/	JSON annotations (bbox, seg, Likert)
metadata.json	Vendor, model, acquisition parameters
splits/train.txt, val.txt, test.txt	Patient ID lists
schema/dataset_schema.md	Canonical record schema
schema/dicom_cleaning_spec.md	De-identification protocol
checksums.sha256	SHA-256 hash of every file
LICENSE.txt	Release license (PhysioNet credentialed)

Table 2: Data record layout. Per-patient directories contain all paired acquisitions and the full per-patient annotation set; top-level files enumerate splits, schema, and integrity hashes.

Aggregate counts are summarized in Table 3.

Cohort	Patients	Vendors	Sites	Anatomy
Training	[TODO: N]	[TODO: N]	[TODO: N]	chest, abdomen
Validation	[TODO: N]	[TODO: N]	[TODO: N]	chest, abdomen
Test	[TODO: N]	[TODO: N]	[TODO: N]	chest, abdomen
Total	[TODO: N]	[TODO: N]	[TODO: N]	—

Table 3: Cohort composition. Final patient counts are filled at submission.

Technical Validation

We performed four classes of technical validation:

Reconstruction sanity

For every full-dose scan we recomputed FBP from the released sinogram and verified that the recomputed image matches the released reconstruction within $\leq$ [TODO: FP tolerance] HU at every voxel. This protects against an off-by-one indexing error between sinogram release and image release, which is a common silent failure mode of CT-dataset releases.

Real- vs simulated-low-dose comparison

On the [TODO: N_{vendor A}] patients with paired real-low-dose acquisitions, we measured the per-patient *distributional* agreement between the simulated 25%-dose image (generated from the patient’s own full-dose sinogram) and the real 25%-dose acquisition. The agreement metrics are reported in Figure 1. The purpose is not to assert simulation equivalence (it is well known that simulated low-dose noise underestimates real low-dose noise in some regimes (Wang et al., 2020)) but to make the discrepancy visible so users can decide which split is appropriate for their evaluation.

[TODO: FIGURE: Per-patient distributional agreement between simulated 25%-dose and real 25%-dose acquisitions on paired-acquisition patients]

Figure 1: Per-patient distributional agreement between simulated 25%-dose and real 25%-dose acquisitions on paired-acquisition patients (placeholder; populate when data lands).

Inter-rater annotation reliability

Cohen’s κ for lung nodule detection (per-radiologist, per-scan) and Dice for liver lesion segmentation are summarized in Table 4. We report agreement broken down by dose level so that any tendency of low-dose images to depress annotator agreement is visible.

[TODO: TABLE: Inter-rater annotation reliability (κ for nodule detection; Dice for liver lesion segmentation), broken down by dose level]

Table 4: Inter-rater annotation reliability (κ for nodule detection; Dice for liver lesion segmentation), broken down by dose level (placeholder; populate when data lands).

Baseline reproductions

We trained four community-standard baselines on the PWM-LDCT training split and report their test-split performance in Table 5. The baselines are RED-CNN (Chen et al., 2017), a transformer-based method ([TODO: select]), a diffusion-based method ([TODO: select]), and the PWM reference unrolled iterative reconstruction (to be confirmed, 2027b). The objective of this section is not to declare a winner but to (i) confirm that all baselines successfully train and evaluate on the released dataset using the provided loader, and (ii) provide initial seed entries for the permanent leaderboard.

5-tuple credentials per signal-equivalence framework

For each baseline we report the corresponding 5-tuple signal-equivalence credential (to be confirmed, 2027a) for the canonical lung nodule detection task at 25% dose. The credentials are summarized in

[TODO: TABLE: Test-split PSNR / SSIM / LPIPS / task-AUC for four reproduced baselines on PWM-LDCT]

Table 5: Test-split PSNR / SSIM / LPIPS / task-AUC for four reproduced baselines on PWM-LDCT (placeholder; populate when data lands).

177 Table 6; the per-method credential JSON is provided alongside each baseline in the release.

[TODO: TABLE: 5-tuple signal-equivalence credentials per baseline at the canonical lung-nodule detection task, 25% dose]

Table 6: 5-tuple signal-equivalence credentials per baseline at the canonical lung-nodule detection task, 25% dose (placeholder; populate when data lands).

178 Usage Notes

179 Loading the data

180 Install the data loader:

```
181 pip install pwm_ldct_loader
```

182 Load a split:

```
183 from pwm_ldct_loader import LowDoseCTDataset
184 train = LowDoseCTDataset(root="/path/to/pwm_ldct",
185                           split="train", seed=42)
186 sample = train[0]
187 # sample["full_dose"]    -> Tensor [H, W]
188 # sample["low_dose"]     -> Tensor [H, W] (real or sim)
189 # sample["sinogram"]     -> Tensor [views, dets]
190 # sample["annotations"] -> dict
191 # sample["metadata"]     -> dict
```

192 The loader pins random seeds, propagates them to PyTorch / NumPy / CUDA RNGs, and uses
193 deterministic CuDNN settings.

194 Reproducing the baseline results

```
195 docker run --gpus all \
196     -v /path/to/pwm_ldct:/data \
197     -v $(pwd)/output:/output \
198     pwm-ldct-red-cnn:v1 \
199     eval --dataset /data --split test \
200         --out /output/results.json --seed 42
```

201 The output `results.json` should match the published values bit-identically (within IEEE 754
202 FP tolerance) on equivalent hardware.

Submitting to the permanent leaderboard

Submissions to the PWM Low-Dose CT Challenge leaderboard ([to be confirmed, 2028](#)) take the form of a PWM RunBundle (Docker image + eval entry point + `results.json` + 5-tuple credentials ([to be confirmed, 2027a](#))). The submission contract is documented at the leaderboard portal.

Cross-vendor evaluation

A method developer can request a held-out-vendor split via the loader’s `leave_vendor_out=vendor` argument; this filters the training split to exclude the named vendor and reassigns its patients to a separate evaluation slice. This is how we measure cross-vendor generalization in the reference method paper ([to be confirmed, 2027b](#)).

Reuse cases

We anticipate four primary reuse modes:

1. **Reconstruction-method benchmarking.** Train on the released training split, evaluate on the released test split, submit a 5-tuple credential under the signal-equivalence framework ([to be confirmed, 2027a](#)), and optionally post the verified score to the public leaderboard ([to be confirmed, 2028](#)).
2. **Cross-vendor robustness studies.** Use the leave-one-vendor-out evaluation API to characterize how a method generalizes from a training-vendor distribution to a held-out-vendor distribution.
3. **Simulation-vs-real noise modeling.** Use the paired real-low-dose and simulated-low-dose images on the **[TODO: $N_{\text{vendor A}}$]** patients with real paired acquisitions to characterize the discrepancy between common Poisson-thinning simulators and physical low-dose acquisition — a long-standing gap in the field that this dataset’s pairing structure now lets researchers quantify.
4. **Downstream-task analysis.** The per-scan diagnostic-quality Likert labels and per-lesion annotations support studies that correlate computational reconstruction quality with radiologist-perceived diagnostic utility, beyond pixel-wise PSNR / SSIM.

Code Availability

- Dataset loader: https://pypi.org/project/pwm_ldct_loader/ (Apache 2.0)
- Preprocessing Dockerfiles: https://github.com/integritynoble/low_dose_CT/tree/main/WS-1_dataset/pipelines/
- Forward model used for low-dose simulation: https://github.com/integritynoble/pwm/tree/main/packages/pwm_core/contrib/modalities/ct_radon.py
- On-chain L3 spec hash: `pwm_l3_spec:pwm_low_dose_ct_benchmark:1.0.0` on PWM main-net at **[TODO: contract address]**.

Author contributions

[TODO: Fill at submission. Use the CRediT taxonomy: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing-Original Draft, Writing-Review & Editing, Visualization, Supervision, Project administration, Funding acquisition.]

Competing interests

[TODO: Declare at submission.]

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